

Chhaileng Taing

Embedded Systems Engineer

📍 France ✉ chhaileng.taing@protonmail.com 🔗 ChhailengTAING

ABOUT ME

I am an Electronics and Embedded Systems Engineer with practical experience in embedded software, communication protocols, and computer vision, acquired through various projects completed during my studies. In addition to my analytical and collaborative nature, I speak and write fluently in Chinese and English, which allows me to thrive in competitive and international environments. I am eager to build my expertise with projects related to robotics, medical imaging, telecommunication systems, and space applications.

TECHNICAL SKILLS

Programming Languages: C/C++, Python, MATLAB

FPGA & DSP: VHDL, Xilinx Vivado, HLS, RTL simulation, FIR/IIR filters, FFT/IFFT

Embedded Platforms & RTOS: STM32, ESP32, Arduino, FreeRTOS, Linux, ROS

Communication Protocols: UART, I2C, SPI, nRF24L01, LoRa, MQTT, HTTP

AI & Computer Vision: PyTorch, OpenCV

Test, Debug & Validation: Oscilloscope, Logic Analyzer, JTAG

EDA & Simulation: ADS, HFSS, KiCad, OrCAD, LTspice, Simulink

Tools: Git, Bash

LANGUAGES

English: Fluent (Written and Spoken)

Chinese (Mandarin): Fluent (Written and Spoken)

Khmer: Native

French: Intermediate, professionally capable

EDUCATION

MSc	Toulouse INP ENSEEIHT , Systems & Electronic Engineering	Toulouse France
	- Electronic Systems for Embedded and Communicating Applications Program - Courses in Embedded Systems, Digital/Analog Electronics, Signal and Image processing, Telecoms, Radar and Remote Sensing - Research project on Machine Learning applied to Embedded Systems.	Sept 2024 – Oct 2025
B.Eng	Institut de Technologie du Cambodge , Electrical and Energy Engineering	Phnom Penh Cambodia
	- Engineer's Degree (Bac+5) - Coursework in Electronics, Control Systems, and Automation. - GPA: 4.00/4.00	Oct 2018 – July 2023

PROFESSIONAL EXPERIENCE

LAAS CNRS , Embedded Computer Vision Engineer (M2 Internship Research Contract)	Toulouse France
Development of a Keypoint Matching system for the vision system of an agricultural robot	Apr 2025 – Sept 2025
- Built an dataset of over 20 000 segmented images. - Run fine-tuning and evaluation of Glue Factory models for keypoint matching pipeline. - Automated the full experiment workflow with Python and Bash scripts on Linux (dataset segmentation, training launches, metric extraction). - Worked on multi-view 3D reconstruction using COLMAP for noise-less depth perception	
Sirindhorn International Institute of Technology , Internship in ML/Computer Vision	Thailand / Cambodia
Development of an automated detection and quality control system for industrial shoe production.	Oct 2023 – Jan 2024
- Trained a YOLOv8 model for shoe detection on the production line. - Built an OpenCV pipeline for total counting and quality classification.	

Institut de Technologie du Cambodge (ITC), Teaching Assistant – Digital Electronics

- Conducted laboratory sessions on digital electronics.
- Supervised LTspice simulations, circuit design, and laboratory safety.

Phnom Penh
Cambodia
Sept 2023 – Aug 2024

Automation Laboratory, ITC, Embedded Software Research Assistant

Development of a robot for the ABU Robocon competition

- Developed the embedded firmware in **C**, implementing sensor acquisition, motor control, and wireless communication.
- Designed the **3D mapping subsystem in Python/C++** for real-time robot navigation.
- Integrated communication using **UART, SPI and I2C** interfaces between modules.

Phnom Penh
Cambodia
Sept 2021 – Aug 2024

KEY TECHNICAL PROJECTS

Real-Time Spectrum Analyzer on FPGA | Thales-sponsored project, ENSEEIHT

Design and validation of a complete DSP chain in VHDL on FPGA.

- Implemented FIR and IIR filtering blocks, with comparison and trade-off analysis.
- Integrated a CORDIC IP block. with validation through RTL simulation in Xilinx Vivado.

ENSEEIHT Toulouse
Oct 2024 – Feb 2025

Bistatic Indoor-Detection Radar System | ENSEEIHT

Design of a low-cost bistatic radar system for full-room indoor motion detection

- Integrated the RF front-end (LNA, mixer, antennas).
- Simulated and validated the system performance in ADS and HFSS.
- Run experimental calibration tests in real conditions.

ENSEEIHT Toulouse
Oct 2024 – Jan 2025

Connected Rain Gauge | Mekong River Commission Competition

Delivered a low-cost connected rain gauge that won 1st prize against teams from across Southeast Asia | Rainfall Category

- Designed and built a complete embedded solution in **C**.
- Developed the communication interface between the device and the cloud with **Lora** and **MQTT**.
- Used **FreeRTOS** for real-time data acquisition.
- Validated long-term system reliability through continuous endurance testing, by writing Python and Bash test scripts on Linux to log, parse, and verify telemetry data against expected sensor behavior.

Laos
Dec 2022 – Apr 2023

REFERENCES

Dr Adam Quotb

Responsable du master ESECA

- Email : adam.quotb@toulouse-inp.fr

Dr. Romane Scherrer

Postdoctoral Researcher, RAP team, LAAS-CNRS

- Internship Supervisor
- Email: romane.scherrer@laas.fr